

Integral Calculus and Differential Equations

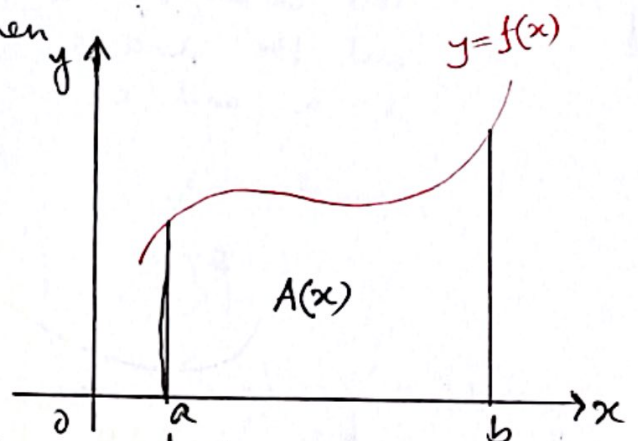
Lecture 1:

AREA and DERIVATIVES

Isaac Newton and Gottfried Leibniz independently discovered a fundamental relationship between areas and derivatives.

They showed that if f is a nonnegative continuous function on the interval $[a, b]$, and if $A(x)$ denotes the area under the graph of f over the interval $[a, x]$ where x is any point in the interval $[a, b]$, then

$$A'(x) = f(x).$$



If we increase x by a small amount Δx ,

the change in the area is approximately the height of the function $f(x)$ times the small width Δx . i.e.,

$$\Delta A \approx f(x) \cdot \Delta x$$

To get the exact rate of change, we take $\Delta x \rightarrow 0$,

$$\frac{dA}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta A}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{f(x) \cdot \Delta x}{\Delta x} = f(x)$$

Which means,

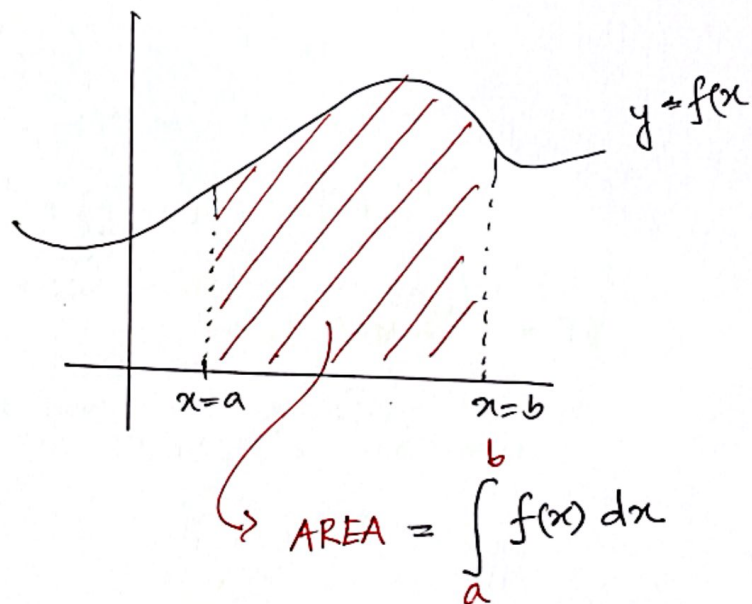
$$\boxed{A'(x) = f(x)}$$

Definite Integral

Given a curve : $y = f(x)$.

Notation : $\int_a^b f(x) dx$

this calculates the "AREA" between the curve $y = f(x)$ and the ~~ax~~ x-axis, between the values of $x = a$ and $x = b$.



Okay, so how do we actually calculate this ?

We'll start with two approaches :

(i) Using GEOMETRY

(ii) APPROXIMATING / ESTIMATING the area using a finite number of RECTANGLES .

S.1 - Areas | Estimating Areas with Finite Sums

ex: (1) Calculate the area between the curve
 $y = \sqrt{49 - x^2} + 1$ and the x-axis,
between $x = -7$ and $x = 7$

i.e. Calculate the value of: $\int_{-7}^7 (\sqrt{49 - x^2} + 1) dx$

we'll do this by DRAWING the curve

$$y = \sqrt{49 - x^2} + 1$$

$$y - 1 = \sqrt{49 - x^2}$$

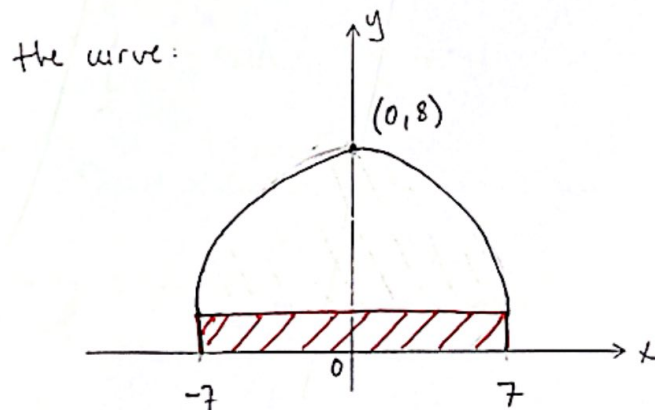
$$(y - 1)^2 = 49 - x^2$$

$$x^2 + (y - 1)^2 = 49$$

$$(x - 0)^2 + (y - 1)^2 = (7)^2$$

this is a CIRCLE CENTERED at $(x, y) = (0, 1)$
with RADIUS $r = 7$

Since the curve has $y = \oplus \sqrt{\dots}$
we draw the TOP HALF of the circle.



the area is a semicircle of radius 7, & a rectangle.

$$\frac{1}{2} \pi (7)^2 + 14(1) = \frac{49\pi}{2} + 14$$

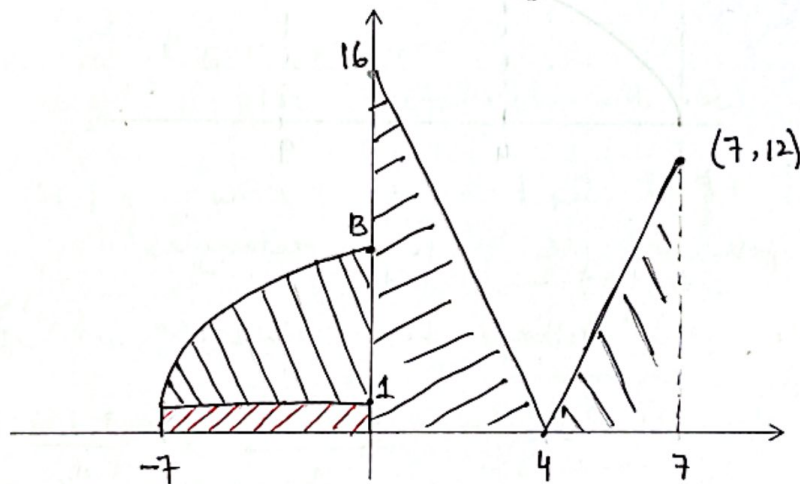
NOTE: on midterms & finals you may see:

$$\text{let } f(x) = \begin{cases} \sqrt{49-x^2} + 1 & \text{if } x \leq 0 \\ |16-4x| & \text{if } x > 0 \end{cases}$$

and calculate: $\int_{-7}^7 f(x) \cdot dx$

Draw the curve $y = \sqrt{49-x^2} + 1$
only for x 's between -7 & 0

and: $y = |16-4x|$
only for x 's between 0 & 7



$$\begin{aligned} \int_{-7}^7 f(x) dx &= 7(1) + \frac{1}{4} \pi (7)^2 + \frac{1}{2} (4)(16) + \frac{1}{2} (3)(12) \\ &= \underline{\underline{\frac{49\pi}{4} + 57}} \end{aligned}$$

if you are asked to use GEOMETRY to calculate a definite integral, it's because the area is made up of:

- circles (or semi-circles)
- rectangles
- triangles

EXTRA PROBLEMS

1. Evaluate the integrals using appropriate formulas from geometry.

$$(i) \int_{-1}^3 (4-5x) dx$$

$$(ii) \int_{-2}^2 (1-3|x|) dx$$

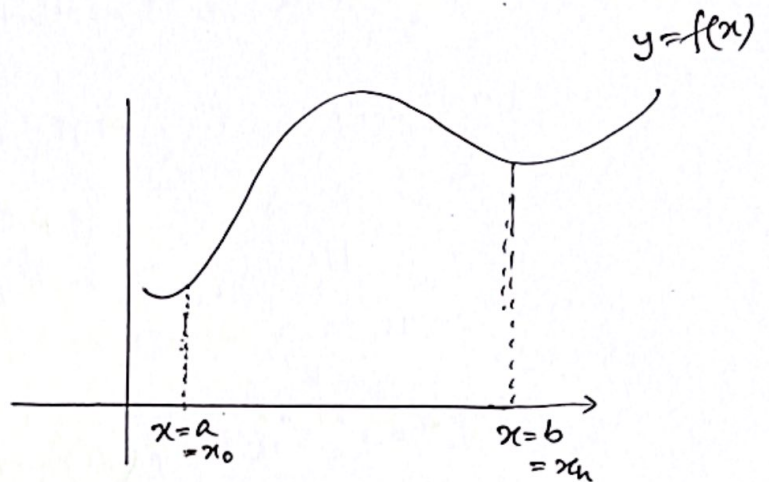
$$(iii) \int_0^1 (x+2\sqrt{1-x^2}) dx$$

$$(iv) \int_{-3}^0 (2+\sqrt{9-x^2}) dx$$

Lecture 11

APPROXIMATE the area using a finite number of RECTANGLES

Given a curve : $y = f(x)$



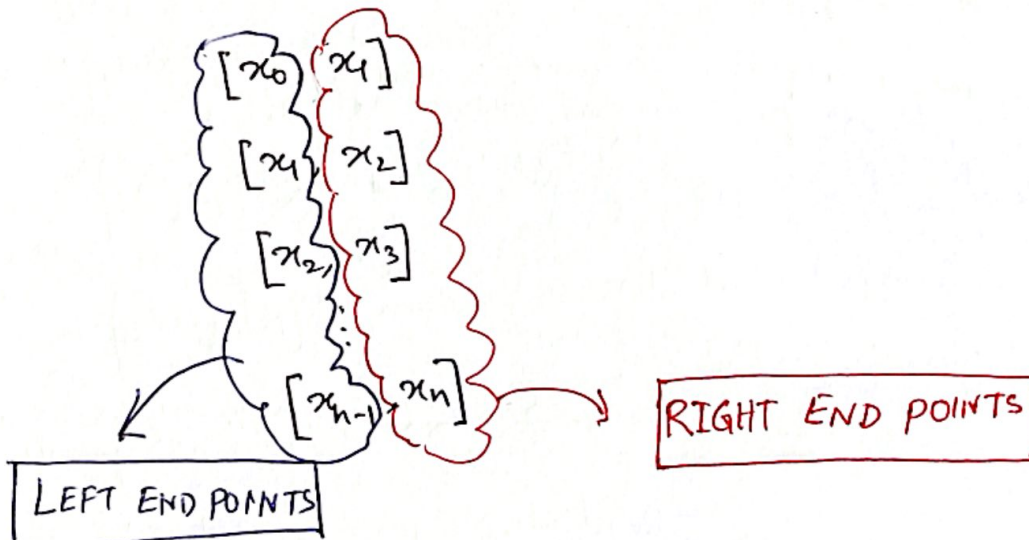
Suppose we use "n" rectangles.

Then each rectangle has a width we'll call Δx .

$$\Delta x = \frac{b-a}{n} = \frac{x_n - a}{n}$$

$$x_n = a + n \cdot \Delta x$$

We have n "sub-intervals" between $x = a$ & $x = b$.



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Here I'll use the RIGHT ENDPOINTS.

In each sub-interval we measure the AREA of a RECTANGLE:

$$A = \text{HEIGHT} \times \text{WIDTH}$$

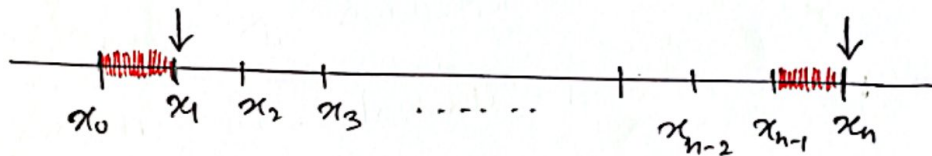
$$A_i = f(x_i) \cdot \Delta x$$

When using the RIGHT ENDPOINTS, the expression:

$$A \approx \sum_{i=1}^n f(x_i) \Delta x = f(x_1) \Delta x + f(x_2) \Delta x + f(x_3) \Delta x + \dots + f(x_n) \Delta x$$

is called a RIGHT RIEMANN Sum: R_n

Here $x_i = a + i \Delta x$



x_1 is the RIGHT ENDPOINT of the 1st interval
 x_2 " " - - - - - 2nd interval
 x_3 - - - - - 3rd - - -
 $\vdots$
 x_n _____ hth "

Le.

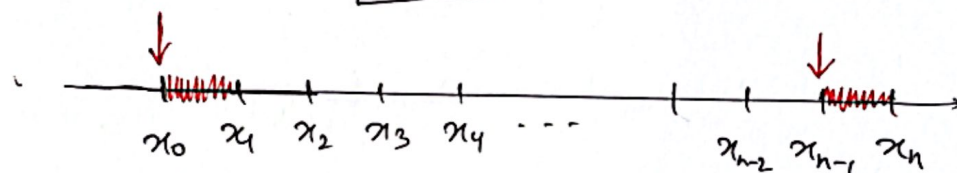
When using the LEFT ENDPOINTS, the expression:

$$A \approx \sum_{i=0}^{n-1} f(x_i) \Delta x = \overbrace{f(x_0) \Delta x}^{\Delta} + f(x_1) \Delta x + \dots + f(x_{n-1}) \Delta x$$

$$= \sum_{i=1}^n f(x_{i-1}) \Delta x$$

is called a Left RIEMANN Sum = L_n

Here $x_{i-1} = a + (i-1) \Delta x$



x_0 is the LEFT ENDPOINT of the 1st interval

x_1 2nd "

x_2 ———— 3rd "

$\vdots$

x_{n-1} ———— n^{th} "

Some Important finite sum:

1. $\sum_{i=1}^n i = \frac{n(n+1)}{2}$

2. $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$

3. $\sum_{i=1}^n i^3 = \left[\frac{n(n+1)}{2} \right]^2$

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EXTRA PROBLEMS

Given the following functions on an interval. Find Riemann sum R_n and also $\lim_{n \rightarrow \infty} R_n$.

1. $f(x) = x^2 - 3x + 2$ on $[0, 2]$

2. $f(x) = 5 - x^3$ on $[0, 4]$

D.

Book: THOMAS' Calculus, EARLY TRANSCENDENTALS,

- G. B. THOMAS

PAGE: 317

- J. HASS

PROBLEMS: 43-50.

- C. HEIL

- M. D. WEIR

1

Extra Problems

Book: Calculus: Early Transcendental BY H. Anton

PAGE: 351:

PROBLEMS: 35-48

• "The Riemann Integral" •

2 The Riemann Integral

The idea behind the Riemann integral is to use very simple approximations for the area of A . Taking a better and better approximations, we can say that "in the limit", we get exactly the area of A under the curve.

Cauchy's work on definite integrals, particularly in his 1823 publication (Calcul Infinitésimal), introduced the concept of partitioning the interval of integration which is a fundamental element of the modern definition of the definite integral.

$$A = \sum_{i=1}^n f(x_{i-1}) \cdot (x_i - x_{i-1})$$

Let $[a, b]$ be a given interval. A partition P of $[a, b]$ is a finite set of points $x_0, x_1, x_2, \dots, x_{n-1}, x_n$ such that $a = x_0 < x_1 < x_2 < \dots < x_{n-1} < x_n = b$ and we write $P = \{x_0, x_1, x_2, \dots, x_{n-1}, x_n\}$. Riemann(1851) represented integral by using the upper sums corresponding to a partition $P = \{a = x_0, x_1, x_2, \dots, x_{n-1}, x_n = b\}$.

Throughout this and next some lectures (before Improper Integral) we assume that f is a bounded function on $[a, b]$. For the given partition P of $[a, b]$, we define

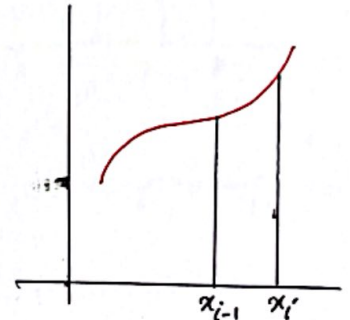
$$M_i = \sup_{a \leq x \leq b} f(x) = \sup\{f(x) : x_{i-1} \leq x \leq x_i\}$$

$$m_i = \inf_{a \leq x \leq b} f(x) = \inf\{f(x) : x_{i-1} \leq x \leq x_i\}$$

and,

$$U(P, f) = \sum_{i=1}^n M_i(x_i - x_{i-1})$$

$$L(P, f) = \sum_{i=1}^n m_i(x_i - x_{i-1})$$



The last two equations are called **upper Darboux sum** and **lower Darboux sum** respectively of f with respect to P .

f is integrable $\iff \inf U(P, f) = \sup L(P, f)$. And then this is called the Riemann integral.

Definition

Let f be a real-valued function which is bounded on $[a, b]$.

The upper integral of f on $[a, b]$ is defined by

$$\int_a^b f(x) dx = \inf_P U(P, f)$$

The lower integral of f on $[a, b]$ is defined by

$$\int_a^b f(x) dx = \sup_P L(P, f)$$

Riemann-integrable:

Let f be a real-valued function on $[a, b]$. We say that f is Riemann-integrable on $[a, b]$ if f is bounded on $[a, b]$,

and

$$\int_a^b f(x) dx = \int_a^b f(x) dx$$

And we define the integral of f on $[a, b]$,

$$\int_a^b f(x) dx = \int_a^b f(x) dx = \int_a^b f(x) dx$$

□

Theorem: If f is continuous on $[a, b]$ then f is integrable.

Proof:

~~Suppose f is continuous at a , $\exists \delta > 0$ such that f is bounded above on $(a - \delta, a + \delta)$.~~

Before starting the proof of above Theorem, we want to state a lemma.

Lemma: Let $a < b$ and $f: [a, b] \rightarrow \mathbb{R}$ be continuous. Then for every $\epsilon > 0 \exists \delta > 0$ such that
 $x, y \in [a, b]$ and $|x - y| < \delta \Rightarrow |f(x) - f(y)| < \epsilon$.

[This proof will be provided on Real Analysis I course]

Now proof for Theorem:

Let $\epsilon > 0$. Using the above lemma, choose $\delta > 0$ such that

$$|f(x) - f(y)| < \epsilon \text{ whenever } x, y \in [a, b] \text{ and } |x - y| < \delta.$$

Let $P = \{a = x_0, x_1, x_2, \dots, x_n = b\}$ be a partition of $[a, b]$ such that $\Delta x_i < \delta$ for all $i = 1, 2, \dots, n$.

Then, there exists $x_i^*, \xi_i^* \in [x_{i-1}, x_i]$ such that $f(x_i^*) = M_i$

and $f(\xi_i^*) = m_i \quad \forall i = 1, 2, \dots, n$.

Therefore, $M_i - m_i \leq \epsilon \quad \forall i = 1, 2, \dots, n$

$$\text{Hence, } U(P, f) - L(P, f) = \sum_{i=1}^n (M_i - m_i) \Delta x_i \leq \epsilon (b - a)$$

which implies that f is integrable.

□

Theorem: [Riemann's criterion for integrability]

Let $f: [a, b] \rightarrow \mathbb{R}$ be a bounded function. Then f is integrable if and only if for every $\epsilon > 0$ there exists a partition P such that $U(P, f) - L(P, f) < \epsilon$. ————— ①

Proof: Suppose that $U(P, f) - L(P, f) < \epsilon$ holds.

Let $\epsilon > 0$ and P satisfy ①. Then

$$L(P, f) \leq \int_a^b f(x) dx \leq \int_a^b f(x) dx \leq U(P, f)$$

Here $L(P, f) \leq \int_a^b f(x) dx$

and $\int_a^b f(x) dx \leq U(P, f)$

Therefore, $\int_a^b f(x) dx - \int_a^b f(x) dx \leq U(P, f) - L(P, f) < \epsilon$ From ①

i.e., $\int_a^b f(x) dx - \int_a^b f(x) dx < \epsilon$.

Conversely, suppose f is integrable and $\epsilon > 0$. Then there exist partitions P_1 and P_2 such that

$$L(P_1, f) > \int_a^b f(x) dx - \frac{\epsilon}{2} \text{ and } U(P_2, f) < \int_a^b f(x) dx + \frac{\epsilon}{2}$$

Let P be the common refinement of P_1 and P_2 . Then

$$\int_a^b f(x) dx - \frac{\epsilon}{2} < L(P_1, f) \leq L(P, f) \leq U(P, f) \leq U(P_2, f) < \int_a^b f(x) dx + \frac{\epsilon}{2}$$

$\therefore U(P, f) - L(P, f) < \epsilon$. ■

Definition Riemann Integral

A bounded function $f: [a, b] \rightarrow \mathbb{R}$ is said to be Riemann integrable or integrable on $[a, b]$ if

$$\int_a^b f(x) dx = \int_a^b f(x) dx.$$

† Then $\int_a^b f(x) dx$ is called the Riemann integral of f .

Example: Consider the function f defined by

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}.$$

Does this function $f(x)$ integrable?

Sol: Consider an interval $[a, b]$, where f is bounded on $[a, b]$.

For any partition P of $[a, b]$,

$$U(P, f) = b - a \text{ and}$$

$$L(P, f) = 0.$$

$$\text{Hence } \int_a^b f(x) dx = 1 \quad \& \quad \int_a^b f(x) dx = 0$$

Therefore f is not integrable over any interval $[a, b]$.

Definition A partition P_2 of $[a, b]$ is said to be finer than a partition P_1 if $P_2 \supset P_1$. In this case we say that P_2 is a refinement of P_1 .

Theorem: Let P_2 be a refinement of P_1 , then $L(P_1, f) \leq L(P_2, f)$ and $U(P_2, f) \leq U(P_1, f)$.

Proof: We first assume that P_2 contains just one more point than P_1 . Let this extra point be x^* .

Suppose $x_{i-1} < x^* < x_i$, where x_{i-1} and x_i are consecutive points of P_1 .

Let $w_1 = \inf \{ f(x) : x_{i-1} \leq x \leq x^* \}$ and

$$w_2 = \inf \{ f(x) : x^* \leq x \leq x_i \}$$

Then $w_1 \geq m_i$ and $w_2 \geq m_i$ where $m_i = \inf \{ f(x) : x_{i-1} \leq x \leq x_i \}$

~~$$\text{Then } L(P_2, f) - L(P_1, f) = w_1(x^* - x_{i-1}) -$$~~

$$\text{Now, } L(P_1, f) = m_i(x_i - x_{i-1})$$

$$\text{and } L(P_2, f) = w_1(x^* - x_{i-1}) + w_2(x_i - x^*)$$

$$\begin{aligned} \therefore L(P_2, f) - L(P_1, f) &= w_1(x^* - x_{i-1}) + w_2(x_i - x^*) - m_i(x_i - x_{i-1}) \\ &= w_1(x^* - x_{i-1}) - m_i(x^* - x_{i-1}) + w_2(x_i - x^*) - m_i(x_i - x^*) \\ &= (w_1 - m_i)(x^* - x_{i-1}) + (w_2 - m_i)(x_i - x^*) \\ &\geq 0 \end{aligned}$$

$$\begin{aligned} \text{since } w_1 - m_i &\geq 0 \\ w_2 - m_i &\geq 0 \end{aligned}$$

Therefore $L(P_2, f) \geq L(P_1, f)$. 6